SUMS OF POWERS VIA CENTRAL FINITE DIFFERENCES AND NEWTON'S FORMULA

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Abstract. In this manuscript, we derive closed formulas for multifold sums of powers of integers by combining the central Newton interpolation formula with hockey-stick identities for binomial coefficients. We further provide Wolfram Mathematica programs for the efficient verification of the derived identities.

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1. Auxiliary Lemmas

This section introduces the lemmas, definitions, and conventions used in the main results. We use Donald Knuth's notation for *multifold sums of powers* as given in [1] because it is concise.

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Definition 1.1 (Multifold sums of powers)**.** *For non-negative integers r, n, and m, define the* multifold sums of powers *recursively by*

$$\Sigma^0 n^m = n^m,$$

$$\Sigma^1 n^m = \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,$$

$$\Sigma^{r+1} n^m = \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.$$

We define the falling factorial notation as follows.

Definition 1.2 (Falling factorials)**.** *For integers n and k, define the* falling factorial *by*

$$(n)_k = \begin{cases} 1, & \text{if } k = 0, \\ n(n-1)(n-2)\cdots(n-k+1) = \prod_{j=0}^{k-1}(n-j), & \text{if } k > 0, \\ 0, & \text{otherwise.} \end{cases}$$

We define the central factorial notation as follows.

Definition 1.3 (Central factorials)**.** *For integers n and k, define the* central factorial *by*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ n\left(n+\frac{k}{2}-1\right)\left(n+\frac{k}{2}-2\right)\cdots\left(n-\frac{k}{2}+1\right) = n\left(n+\frac{k}{2}-1\right)_{k-1}, & \text{if } k > 0, \end{cases}$$

where $\left(n+\frac{k}{2}-1\right)_{k-1} = \prod_{j=0}^{k-2}\left(n+\frac{k}{2}-1-j\right)$ is a falling factorial.

For further discussion of generalized factorials, see J. F. Steffensen's treatment in [2].

Lemma 1.4 (Central Newton's formula)**.** *Let f be a function defined on the integers, and let x and t be arbitrary integers.*

$$f(x) = \sum_{k=0}^{\infty} \frac{(x-t)^{[k]}}{k!} \delta^k f(t).$$

The kth central finite difference is defined by

$$\delta^k f(t) = \sum_{j=0}^{k} (-1)^j \binom{k}{j} f\left(t + \tfrac{k}{2} - j\right),$$

and $(x-t)^{[k]}$ *denotes the central factorial defined in Definition* (1.3).

Proof. Proofs of the central Newton formula are given in [3, p. 462], in [4, Section 2, Equation (19)], and in [5, Section 6, Equation (21)]. □

The specialization to powers gives the following lemma.

Lemma 1.5 (Newton's formula for powers)**.** *Let* n *and* m *be nonnegative integers, and let* t *be an arbitrary integer.*

$$n^m = \sum_{k=0}^{m} \frac{(n-t)^{[k]}}{k!} \delta^k t^m.$$

Proof. This is the special case of Lemma (1.4) with $f(n) = n^m$. □

The central factorial identity has the following binomial form.

Lemma 1.6 (Central factorials binomial form)**.** *For integers* n *and nonzero* k*,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1}.$$

Proof. We have

$$\frac{n^{[k]}}{k!} = \frac{n}{k!} \left(n + \frac{k}{2} - 1\right)_{k-1} = \frac{n}{k(k-1)!} \left(n + \frac{k}{2} - 1\right)_{k-1} = \frac{n}{k} \tbinom{n+\frac{k}{2}-1}{k-1},$$

by the falling-factorial identity $\frac{(x)_n}{n!} = \binom{x}{n}$. □

The following identity separates a factor of n from a binomial coefficient.

Lemma 1.7 (Binomial decomposition)**.** *Let* r*,* n*, and* m *be nonnegative integers.*

$$n \binom{n+r}{m} = (m+1) \binom{n+r}{m+1} - (r-m) \binom{n+r}{m}.$$

Proof. Expanding the right-hand side gives

$$n\tbinom{n+r}{m} = m\tbinom{n+r}{m+1} + \tbinom{n+r}{m+1} - r\tbinom{n+r}{m} + m\tbinom{n+r}{m}.$$

Recall the extraction property of binomial coefficients $\tbinom{n}{k+1} = \frac{n-k}{k+1}\tbinom{n}{k}$. The extraction property then gives

$$\tbinom{n+r}{m+1} = \frac{n+r-m}{m+1}\tbinom{n+r}{m}.$$

Thus,

$$n\tbinom{n+r}{m} = m\frac{n+r-m}{m+1}\tbinom{n+r}{m} + \frac{n+r-m}{m+1}\tbinom{n+r}{m} - r\tbinom{n+r}{m} + m\tbinom{n+r}{m}.$$

Factoring out the binomial coefficient $\tbinom{n+r}{m}$ gives

$$\begin{aligned} n\tbinom{n+r}{m} &= \tbinom{n+r}{m}\left[m\frac{n+r-m}{m+1} + \frac{n+r-m}{m+1} - r + m\right] \\ &= \tbinom{n+r}{m}\left[(m+1)\frac{n+r-m}{m+1} - r + m\right] \\ &= \tbinom{n+r}{m} n. \end{aligned}$$

This completes the proof. □

We also use the generalized hockey-stick identity.

Lemma 1.8 (Generalized hockey-stick identity)**.** *For arbitrary integers a, b, and j,*

$$\sum_{k=a}^{b} \binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

Proof. We have $\sum_{k=a}^{b}\tbinom{k}{j} = \tbinom{a}{j} + \tbinom{a+1}{j} + \cdots + \tbinom{b}{j} = \left(\sum_{k=0}^{b}\tbinom{k}{j}\right) - \left(\sum_{k=0}^{a-1}\tbinom{k}{j}\right)$. The hockey-stick identity $\sum_{k=0}^{n}\tbinom{k}{j} = \tbinom{n+1}{j+1}$ therefore gives

$$\sum_{k=a}^{b}\tbinom{k}{j} = \left(\sum_{k=0}^{b}\tbinom{k}{j}\right) - \left(\sum_{k=0}^{a-1}\tbinom{k}{j}\right) = \tbinom{b+1}{j+1} - \tbinom{a}{j+1}.$$

This completes the proof. □

The centered version is as follows.

Lemma 1.9 (Centered hockey-stick identity)**.** *Let n be a positive integer, and let t, k, and r be arbitrary integers.*

$$\sum_{j=1}^{n} \binom{j - t + \frac{k}{2} + r}{k} = \binom{n - t + \frac{k}{2} + r + 1}{k+1} - \binom{1 - t + \frac{k}{2} + r}{k+1}.$$

Proof. Substituting $a = 1 - t + \frac{k}{2} + r$ and $b = n - t + \frac{k}{2} + r$ into Lemma (1.8) gives

$$\sum_{j=1}^{n} \tbinom{j-t+\frac{k}{2}+r}{k} = \sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r} \tbinom{j}{k} = \tbinom{n-t+\frac{k}{2}+r+1}{k+1} - \tbinom{1-t+\frac{k}{2}+r}{k+1}.$$

This completes the proof. □

The following formula gives the multifold sum of zeroth powers.

Lemma 1.10 (Multifold sums of zero powers)**.** *Let r and n be nonnegative integers.*

$$\Sigma^r\, n^0 = \binom{r + n - 1}{r}.$$

Proof.

(1) When $r = 0$, Definition (1.1) gives $\Sigma^0\, n^0 = 1$.

(2) When $r = 1$, we have $\Sigma^1\, n^0 = \sum_{k=1}^{n} 1 = \binom{n}{1}$.

(3) When $r = 2$, we have $\Sigma^2\, n^0 = \sum_{k=1}^{n} \binom{k}{1} = \binom{n+1}{2}$.

(4) When $r = 3$, we have $\Sigma^3\, n^0 = \sum_{k=1}^{n} \binom{k+1}{2} = \binom{n+2}{3}$.

(5) For general r, the hockey-stick identity gives $\Sigma^r\, n^0 = \sum_{k=1}^{n} \Sigma^{r-1}\, k^0 = \sum_{k=1}^{n} \binom{k+r-2}{r-1} = \binom{r+n-1}{r}$.

This completes the proof. □

Throughout, we use the following convention.

Convention 1.11. *For all x,*

$$x^0 = 1.$$

Donald Knuth discusses the convention $x^0 = 1$ for all x, including zero, in Concrete Mathematics, as described in [6, p. 162] and in his notes in [7].

2. Introduction

In this manuscript, we derive formulas for sums of powers by combining the central Newton interpolation formula for powers in Lemma (1.5) with the hockey-stick identity in Lemma (1.9). We showcase the algorithm for finding closed forms for sums of powers of integers, as proposed in [8, Alg. (11.2)]. More precisely, the main idea is to identify the binomial basis in variants of Newton's formula associated with different difference operators. For example,

$$\begin{cases} f(x) &= \sum_{k=0}^{\infty} \frac{(x-a)^{[k]}}{k!} \delta^k f(a) = \sum_{k=0}^{\infty} \frac{x-a}{k} \binom{x-a+\frac{k}{2}-1}{k-1} \delta^k f(a), \\ f(x) &= \sum_{k=0}^{\infty} \frac{(x-a)_k}{k!} \Delta^k f(a) = \sum_{k=0}^{\infty} \binom{x-a}{k} \Delta^k f(a), \\ f(x) &= \sum_{k=0}^{\infty} \frac{(x-a)^{(k)}}{k!} \nabla^k f(a) = \sum_{k=0}^{\infty} \binom{x-a+k-1}{k} \nabla^k f(a), \\ f(x) &= \sum_{k=0}^{\infty} \frac{(x-a)^k}{k!} \frac{\mathrm{d}^k f(a)}{\mathrm{d}x^k}. \end{cases}$$

Each variant of Newton's formula above has a corresponding binomial basis:

$$\begin{cases} \frac{(x)_n}{n!} &= \frac{1}{n!} x(x-1)(x-2)\cdots(x-n+1) = \binom{x}{n}, \\ \frac{x^{(n)}}{n!} &= \frac{1}{n!} x(x+1)(x+2)\cdots(x+n-1) = \binom{x+n-1}{n}, \\ \frac{x^{[n]}}{n!} &= \frac{1}{n!} \left(n+\frac{k}{2}-1\right)\left(n+\frac{k}{2}-2\right)\cdots\left(n-\frac{k}{2}+1\right) = \frac{x}{n}\binom{x+\frac{n}{2}-1}{n-1}. \end{cases}$$

These bases allow us to combine Newton's formula with a hockey-stick identity to obtain closed forms for sums of powers. Thus, we focus on closed forms for sums of powers using the central Newton formula and the hockey-stick identity in Lemma (1.9).

3. Main Results

We begin with Lemma (1.5), which gives Newton's interpolation formula for powers. Applying Lemma (1.6) to this formula, for non-negative integers n and m and an arbitrary

integer t, gives

$$n^m = \frac{(n-t)^{[0]}}{0!}\delta^0 t^m + \sum_{k=1}^{m} \frac{n-t}{k}\binom{n+t+\frac{k}{2}-1}{k-1}\delta^k t^m$$
$$= t^m + \sum_{k=1}^{m}(n-t)\binom{n-t+\frac{k}{2}-1}{k-1}\frac{\delta^k t^m}{k}.$$

Note that we isolate the term $\frac{(n-t)^{[0]}}{0!}\delta^0 t^m$ to avoid division by zero. By expanding the brackets, we have

$$n^m = t^m + \sum_{k=1}^{m} \frac{\delta^k t^m}{k}\left[n\binom{n-t+\frac{k}{2}-1}{k-1} - t\binom{n-t+\frac{k}{2}-1}{k-1}\right].$$

Notice that the factor $n\binom{n-t+\frac{k}{2}-1}{k-1}$ prevents us from direct use of hockey-stick identity (1.8). The binomial decomposition in Lemma (1.7) handles this factor. Substituting $r = -t+\frac{k}{2}-1$ and $m = k-1$ into Lemma (1.7) gives

$$n\binom{n-t+\frac{k}{2}-1}{k-1} = k\binom{n-t+\frac{k}{2}-1}{k} + \left[t+\frac{k}{2}\right]\binom{n-t+\frac{k}{2}-1}{k-1}.$$

For clarity, applying Lemma (1.7) directly gives

$$n\tbinom{n-t+\frac{k}{2}-1}{k-1} = k\tbinom{n-t+\frac{k}{2}-1}{k-1+1} - \left[-t+\tfrac{k}{2}-1-(k-1)\right]\tbinom{n-t+\frac{k}{2}-1}{k-1}$$
$$= k\tbinom{n-t+\frac{k}{2}-1}{k} - \left[-t-\tfrac{k}{2}\right]\tbinom{n-t+\frac{k}{2}-1}{k-1}$$
$$= k\tbinom{n-t+\frac{k}{2}-1}{k} + \left[t+\tfrac{k}{2}\right]\tbinom{n-t+\frac{k}{2}-1}{k-1}.$$

Thus,

$$n^m = t^m + \sum_{k=1}^{m} \frac{\delta^k t^m}{k}\left[k\binom{n-t+\frac{k}{2}-1}{k} + \left[t+\frac{k}{2}\right]\binom{n-t+\frac{k}{2}-1}{k-1} - t\binom{n-t+\frac{k}{2}-1}{k-1}\right].$$

Combining the terms involving $t\binom{n-t+\frac{k}{2}-1}{k-1}$ and simplifying the factors of k gives

$$n^m = t^m + \sum_{k=1}^{m}\left[\binom{n-t+\frac{k}{2}-1}{k} + \frac{1}{2}\binom{n-t+\frac{k}{2}-1}{k-1}\right]\delta^k t^m.$$

We absorb the term t^m into the sum to obtain

$$n^m = \sum_{k=0}^{m}\left[\binom{n-t+\frac{k}{2}-1}{k} + \frac{1}{2}\binom{n-t+\frac{k}{2}-1}{k-1}\right]\delta^k t^m.$$

We now simplify the binomial basis $\binom{n-t+\frac{k}{2}-1}{k} + \frac{1}{2}\binom{n-t+\frac{k}{2}-1}{k-1}$ to remove the fractional coefficient. Set $a = n - t + \frac{k}{2} - 1$.

$$\binom{a}{k} + \frac{1}{2}\binom{a}{k-1} = \frac{1}{2}\left[2\binom{a}{k} + \binom{a}{k-1}\right] = \frac{1}{2}\left[\binom{a}{k} + \binom{a+1}{k}\right].$$

Therefore,

$$n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\binom{n-t+\frac{k}{2}-1}{k} + \binom{n-t+\frac{k}{2}}{k-1}\right]\delta^k t^m.$$

The upper indices of the binomial coefficients may be fractional, because $\binom{n}{k}$ is a polynomial in n. The identity involving the unsigned Stirling numbers of the first kind $\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]$ makes this explicit:

$$\binom{n}{k} = \sum_{j=0}^{k}\frac{(-1)^{k-j}}{k!}\left[\begin{smallmatrix} k \\ j \end{smallmatrix}\right] n^j.$$

The following example illustrates the identity.

Example 3.1. *For integer* $0 \le t \le 3$*, we have*

$$\begin{aligned}
t = 0: \quad n^3 &= \left(1 + \tbinom{n}{-1}\right)\cdot 0 + \tfrac{1}{2}(1+2n)\cdot\tfrac{1}{4} + \tfrac{1}{2}(2+n+n^2)\cdot 0 \\
&\quad + \tfrac{1}{48}(21 + 46n + 12n^2 + 8n^3)\cdot 6, \\
t = 1: \quad n^3 &= \left(1 + \tbinom{n-1}{-1}\right)\cdot 1 + \tfrac{1}{2}(-1+2n)\cdot\tfrac{13}{4} + \tfrac{1}{2}(2-n+n^2)\cdot 6 \\
&\quad + \tfrac{1}{48}(-21 + 46n - 12n^2 + 8n^3)\cdot 6, \\
t = 2: \quad n^3 &= \left(1 + \tbinom{n-2}{-1}\right)\cdot 8 + \tfrac{1}{2}(-3+2n)\cdot\tfrac{49}{4} + \tfrac{1}{2}(4-3n+n^2)\cdot 12 \\
&\quad + \tfrac{1}{48}(-87 + 94n - 36n^2 + 8n^3)\cdot 6, \\
t = 3: \quad n^3 &= \left(1 + \tbinom{n-3}{-1}\right)\cdot 27 + \tfrac{1}{2}(-5+2n)\cdot\tfrac{109}{4} + \tfrac{1}{2}(8-5n+n^2)\cdot 18 \\
&\quad + \tfrac{1}{48}(-225 + 190n - 60n^2 + 8n^3)\cdot 6.
\end{aligned}$$

We can now apply the centered hockey-stick identity in Lemma (1.9) to obtain a closed formula for sums of powers:

$$\Sigma^1 n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\sum_{j=1}^{n}\binom{j-t+\frac{k}{2}-1}{k}+\sum_{j=1}^{n}\binom{j-t+\frac{k}{2}}{k-1}\right]\delta^k t^m.$$

Applying Lemma (1.9) gives

$$\begin{cases}\sum_{j=1}^{n}\binom{j-t+\frac{k}{2}-1}{k} &= \binom{n-t+\frac{k}{2}+0}{k+1}-\binom{0-t+\frac{k}{2}}{k+1},\\ \sum_{j=1}^{n}\binom{j-t+\frac{k}{2}}{k-1} &= \binom{n-t+\frac{k}{2}+1}{k+1}-\binom{1-t+\frac{k}{2}}{k+1}.\end{cases}$$

We obtain the following closed formula for ordinary sums of powers.

Proposition 3.2 (Ordinary sums of powers)**.** *Let n and m be nonnegative integers, and let t be an arbitrary integer.*

$$\Sigma^1 n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\binom{n-t+\frac{k}{2}+0}{k+1}+\binom{n-t+\frac{k}{2}+1}{k+1}-\binom{0-t+\frac{k}{2}}{k+1}-\binom{1-t+\frac{k}{2}}{k+1}\right]\delta^k t^m.$$

The following example illustrates the formula.

Example 3.3. *For integer $0 \le t \le 4$, we have*

$$\begin{aligned}
t=0: \quad \Sigma^1 n^3 &= 2n\cdot 0+(n+n^2)\cdot\tfrac{1}{4}+\tfrac{1}{6}(n+3n^2+2n^3)\cdot 0\\
&\quad+\tfrac{1}{24}(-n+n^2+4n^3+2n^4)\cdot 6,\\
t=1: \quad \Sigma^1 n^3 &= 2n\cdot 1+(-n+n^2)\cdot\tfrac{13}{4}+\tfrac{1}{6}(n-3n^2+2n^3)\cdot 6\\
&\quad+\tfrac{1}{24}(n+n^2-4n^3+2n^4)\cdot 6,\\
t=2: \quad \Sigma^1 n^3 &= 2n\cdot 8+(-3n+n^2)\cdot\tfrac{49}{4}+\tfrac{1}{6}(13n-9n^2+2n^3)\cdot 12\\
&\quad+\tfrac{1}{24}(-21n+25n^2-12n^3+2n^4)\cdot 6,\\
t=3: \quad \Sigma^1 n^3 &= 2n\cdot 27+(-5n+n^2)\cdot\tfrac{109}{4}+\tfrac{1}{6}(37n-15n^2+2n^3)\cdot 18\\
&\quad+\tfrac{1}{24}(-115n+73n^2-20n^3+2n^4)\cdot 6,\\
t=4: \quad \Sigma^1 n^3 &= 2n\cdot 64+(-7n+n^2)\cdot\tfrac{193}{4}+\tfrac{1}{6}(73n-21n^2+2n^3)\cdot 24\\
&\quad+\tfrac{1}{24}(-329n+145n^2-28n^3+2n^4)\cdot 6.
\end{aligned}$$

Similarly, setting $r = 0$ and $r = 1$ in Lemma (1.9) gives

$$\begin{cases} \sum_{j=1}^{n} \binom{j-t+\frac{k}{2}+0}{k+1} &= \binom{n-t+\frac{k}{2}+1}{k+2} - \binom{1-t+\frac{k}{2}}{k+2}, \\ \sum_{j=1}^{n} \binom{j-t+\frac{k}{2}+1}{k+1} &= \binom{n-t+\frac{k}{2}+2}{k+2} - \binom{2-t+\frac{k}{2}}{k+2}. \end{cases}$$

We can now state the formula for double sums of powers.

Proposition 3.4 (Double sums of powers)**.** *Let n and m be nonnegative integers, and let t be an arbitrary integer.*

$$\begin{aligned} \Sigma^2 n^m = \frac{1}{2} \sum_{k=0}^{m} \Bigg[& \binom{n-t+\frac{k}{2}+1}{k+2} + \binom{n-t+\frac{k}{2}+2}{k+2} \\ & - \left(\binom{0-t+\frac{k}{2}}{k+1} + \binom{1-t+\frac{k}{2}}{k+1} \right) \Sigma^1 n^0 \\ & - \left(\binom{1-t+\frac{k}{2}}{k+2} + \binom{2-t+\frac{k}{2}}{k+2} \right) \Sigma^0 n^0 \Bigg] \delta^k t^m. \end{aligned}$$

The following example illustrates the double-sum formula.

Example 3.5. *For integer $0 \le t \le 2$, we have*

$$\begin{aligned} t = 0: \quad \Sigma^1 n^3 &= \tfrac{1}{2}(n+n^2) \cdot 0 + \tfrac{1}{48}(-3-2n+12n^2+8n^3) \cdot \tfrac{1}{4} \\ &\quad + \tfrac{1}{24}(-2n-n^2+2n^3+n^4) \cdot 0 \\ &\quad + \tfrac{1}{3840}(45+18n-200n^2-80n^3+80n^4+32n^5) \cdot 6, \\ t = 1: \quad \Sigma^1 n^3 &= \tfrac{1}{2}(-n+n^2) \cdot 1 + \tfrac{1}{48}(3-2n-12n^2+8n^3) \cdot \tfrac{13}{4} \\ &\quad + \tfrac{1}{24}(2n-n^2-2n^3+n^4) \cdot 6 \\ &\quad + \tfrac{1}{3840}(-45+18n+200n^2-80n^3-80n^4+32n^5) \cdot 6, \\ t = 2: \quad \Sigma^1 n^3 &= \tfrac{1}{2}(2-3n+n^2) \cdot 8 + \tfrac{1}{48}(-15+46n-36n^2+8n^3) \cdot \tfrac{49}{4} \\ &\quad + \tfrac{1}{24}(-6n+11n^2-6n^3+n^4) \cdot 12 \\ &\quad + \tfrac{1}{3840}(105-142n-360n^2+560n^3-240n^4+32n^5) \cdot 6. \end{aligned}$$

The general case is as follows.

Theorem 3.6 (Multifold sums of powers)**.** *Let r, n, and m be nonnegative integers, and let t be an arbitrary integer.*

$$\Sigma^r n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\binom{n-t+\frac{k}{2}+r-1}{k+r} + \binom{n-t+\frac{k}{2}+r}{k+r} - \sum_{s=0}^{r-1}\left(\binom{s-t+\frac{k}{2}}{k+s+1} + \binom{s-t+\frac{k}{2}+1}{k+s+1}\right)\Sigma^{r-1-s} n^0\right]\delta^k t^m.$$

Lemma (1.10) gives the following formula in binomial form.

Proposition 3.7 (Multifold Binomial sums of powers)**.** *Let r, n, and m be nonnegative integers, and let t be an arbitrary integer.*

$$\Sigma^r n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\binom{n-t+\frac{k}{2}+r-1}{k+r} + \binom{n-t+\frac{k}{2}+r}{k+r} - \sum_{s=0}^{r-1}\left(\binom{s-t+\frac{k}{2}}{k+s+1} + \binom{s-t+\frac{k}{2}+1}{k+s+1}\right)\binom{r-s+n-2}{r-s-1}\right]\delta^k t^m.$$

We have therefore derived closed formulas for multifold sums of powers. The case of forward finite differences Δ and Newton's formula is given in [9]; it follows the algorithm proposed in [8, Alg. (11.2)]. The case of multifold formulas using backward differences ∇ and Newton's formula is given in [10].

4. Remark on Knuth's Formula

Donald Knuth gives a formula for sums of odd powers in terms of the central factorial numbers of the second kind $T(n,k)$ in [1].

Proposition 4.1 (Knuth's odd powers sum)**.** *Let n be a nonnegative integer, and let m be a positive integer.*

$$\Sigma^1 n^{2m-1} = \sum_{k=1}^{m}(2k-1)!\binom{n+k}{2k}T(2m,2k),$$

where $T(2m,2k)$ are central factorial numbers of the second kind.

The sequences A008957, A036969, and A269945 list central factorial numbers of the second kind $T(n,k)$ in the OEIS, as described in [11]. Table 1 lists initial values of $T(2n,2k)$.

n/k	0	1	2	3	4	5	6	7
0	1							
1	0	1						
2	0	1	1					
3	0	1	5	1				
4	0	1	21	14	1			
5	0	1	85	147	30	1		
6	0	1	341	1408	627	55	1	
7	0	1	1365	13013	11440	2002	91	1

Table 1. Central factorial numbers of the second kind $T(2n,2k)$.

It is interesting to notice that Proposition (4.1) directly follows from Central Newton's Formula in Lemma (1.5), evaluated at zero. For $m \geq 1$, the formula gives

$$n^{2m} = \sum_{k=1}^{2m} \frac{n^{[k]}}{k!} \cdot \delta^k 0^{2m}.$$

The term for $k = 0$ vanishes because $\delta^0 0^{2m} = 0$ when $m \geq 1$. For positive integers m and k, the central difference $\delta^k 0^m$ is nonzero exactly when $k \leq m$ and m and k have the same parity:

$$\begin{cases} \delta^k 0^m \neq 0, & \text{when} \quad m \equiv k \pmod 2, \\ \delta^k 0^m = 0, & \text{when} \quad m \not\equiv k \pmod 2, \end{cases}$$

In particular, the difference vanishes when m and k have opposite parity. This parity property applies to powers evaluated at zero; for example, $\delta^3 1^4 = 24 \neq 0$. Since $2m$ is even, the differences $\delta^k 0^{2m}$ vanish for odd k, so we may omit those terms:

$$n^{2m} = \sum_{k=1}^{m} \frac{n^{[2k]}}{(2k)!} \cdot \delta^{2k} 0^{2m}.$$

By the definition of the central factorial numbers of the second kind, we obtain

$$n^{2m} = \sum_{k=1}^{m} n^{[2k]} T(2m, 2k).$$

Therefore,

$$n^{2m} = \sum_{k=1}^{m} (2k)! \frac{n^{[2k]}}{(2k)!} T(2m, 2k).$$

Applying Lemma (1.6) to simplify $\frac{n^{[2k]}}{(2k)!}$ gives

$$n^{2m} = \sum_{k=1}^{m} (2k)! \frac{n}{2k} \binom{n+k-1}{2k-1} T(2m, 2k).$$

Consequently,

$$n^{2m} = \sum_{k=1}^{m} (2k-1)! n \binom{n+k-1}{2k-1} T(2m, 2k).$$

Factoring out n gives

$$n^{2m-1} = \sum_{k=1}^{m} (2k-1)! \binom{n+k-1}{2k-1} T(2m, 2k).$$

Applying the hockey-stick identity $\sum_{k=0}^{n} \binom{k}{j} = \binom{n+1}{j+1}$ gives

$$\Sigma^1 \, n^{2m-1} = \sum_{k=1}^{m} (2k-1)! \binom{n+k}{2k} T(2m, 2k).$$

This formula agrees with Proposition (4.1). Repeated application of the hockey-stick identity $\sum_{k=0}^{n} \binom{k}{j} = \binom{n+1}{j+1}$ generalizes the formula to multifold sums.

Proposition 4.2 (Knuth's odd powers sum multifold)**.** *Let r and n be nonnegative integers, and let m be a positive integer.*

$$\Sigma^{r+1} \, n^{2m-1} = \sum_{k=1}^{m} (2k-1)! \binom{n+k+r}{2k+r} T(2m, 2k),$$

where $T(2m, 2k)$ are central factorial numbers of the second kind.

The following example illustrates Proposition (4.1).

Example 4.3. *For non-negative integers n, and $r = 1$, we have*

$$\Sigma^1\, n^1 = \tbinom{n+1}{2}1,$$

$$\Sigma^1\, n^3 = \tbinom{n+2}{4}6 + \tbinom{n+1}{2}1,$$

$$\Sigma^1\, n^5 = \tbinom{n+3}{6}120 + \tbinom{n+2}{4}30 + \tbinom{n+1}{2}1,$$

$$\Sigma^1\, n^7 = \tbinom{n+4}{8}5040 + \tbinom{n+3}{6}1680 + \tbinom{n+2}{4}126 + \tbinom{n+1}{2}1.$$

The following example illustrates Proposition (4.2).

Example 4.4. *For non-negative integers r, n*

$$\Sigma^r\, n^1 = \tbinom{n+0+r}{1+r}1,$$

$$\Sigma^r\, n^3 = \tbinom{n+1+r}{3+r}6 + \tbinom{n+0+r}{1+r}1,$$

$$\Sigma^r\, n^5 = \tbinom{n+2+r}{5+r}120 + \tbinom{n+1+r}{3+r}30 + \tbinom{n+0+r}{1+r}1,$$

$$\Sigma^r\, n^7 = \tbinom{n+3+r}{7+r}5040 + \tbinom{n+3+r}{5+r}1680 + \tbinom{n+1+r}{3+r}126 + \tbinom{n+0+r}{1+r}1.$$

The coefficients in Proposition (4.2) form the sequence A303675 in the OEIS, as listed in [11]. Proposition (4.2) also motivates the following identity. Combining Newton's formula in Lemma (1.5) with Lemma (1.6) gives

$$n^m = \sum_{k=1}^{m} \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} \delta^k 0^m.$$

Factoring out n gives

$$n^{m-1} = \sum_{k=1}^{m} \frac{1}{k} \binom{n + \frac{k}{2} - 1}{k-1} \delta^k 0^m.$$

Shifting the exponent gives

$$n^m = \sum_{k=1}^{m+1} \frac{1}{k} \binom{n + \frac{k}{2} - 1}{k-1} \delta^k 0^{m+1}.$$

Re-indexing the sum gives

$$n^m = \sum_{k=0}^{m} \frac{1}{k+1} \binom{n + \frac{k}{2} - \frac{1}{2}}{k} \delta^{k+1} 0^{m+1}. \tag{1}$$

Setting $r = 0$ and $t = \frac{1}{2}$ in Lemma (1.9) gives the following proposition.

Proposition 4.5 (Ordinary sums of powers at zero)**.** *Let n and m be nonnegative integers.*

$$\Sigma^1 n^m = \sum_{k=0}^{m} \frac{1}{k+1} \left[\binom{n + \frac{k}{2} + \frac{1}{2}}{k+1} - \binom{\frac{k}{2} + \frac{1}{2}}{k+1} \right] \delta^{k+1} 0^{m+1}.$$

The formula can also be written in terms of central factorial numbers of the second kind:

$$\Sigma^1 n^m = \sum_{k=0}^{m} k! \left[\binom{n + \frac{k}{2} + \frac{1}{2}}{k+1} - \binom{\frac{k}{2} + \frac{1}{2}}{k+1} \right] T(k+1, m+1).$$

Setting $r = 1$ and $t = \frac{1}{2}$ in Lemma (1.9) gives a formula for double sums of powers.

Proposition 4.6 (Double sums of powers at zero)**.** *Let n and m be nonnegative integers.*

$$\Sigma^2 n^m = \sum_{k=0}^{m} \frac{1}{k+1} \left[\binom{n + \frac{k}{2} + \frac{3}{2}}{k+2} - \binom{\frac{k}{2} + \frac{1}{2}}{k+1} \Sigma^1 n^0 - \binom{\frac{k}{2} + \frac{3}{2}}{k+2} \Sigma^0 n^0 \right] \delta^{k+1} 0^{m+1}.$$

The general formula is as follows.

Theorem 4.7 (Multifold sums of powers at zero)**.** *Let r, n, and m be nonnegative integers.*

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[\binom{n + \frac{k}{2} + \frac{2r-1}{2}}{k+r} - \sum_{s=0}^{r-1} \binom{\frac{k}{2} + \frac{2(r-s)-1}{2}}{k+r-s} \Sigma^s n^0 \right] \frac{\delta^{k+1} 0^{m+1}}{k+1}.$$

Proof. By repeatedly applied Lemma (1.9), and Equation (1). □

Combining Lemma (1.10) with Theorem (4.7) gives the following binomial form.

Proposition 4.8 (Multifold sums of powers at zero binomial form)**.** *Let r, n, and m be nonnegative integers.*

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[\binom{n + \frac{k}{2} + \frac{2r-1}{2}}{k+r} - \sum_{s=0}^{r-1} \binom{\frac{k}{2} + \frac{2(r-s)-1}{2}}{k+r-s} \binom{s+n-1}{s} \right] \frac{\delta^{k+1} 0^{m+1}}{k+1}.$$

Proof. By repeatedly applied Lemma (1.9), and Equation (1). □

This formula can also be expressed in terms of central factorial numbers of the second kind:

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[\binom{n + \frac{k}{2} + \frac{2r-1}{2}}{k+r} - \sum_{s=0}^{r-1} \binom{\frac{k}{2} + \frac{2(r-s)-1}{2}}{k+r-s} \binom{s+n-1}{s} \right] k! T(k+1, m+1).$$

The results above follow from evaluating the central Newton formula at zero. Table 2 lists initial values of $T(n,k)$.

n/k	0	1	2	3	4	5	6	7	8
0	1								
1	0	1							
2	0	0	1						
3	0	$\frac{1}{4}$	0	1					
4	0	0	1	0	1				
5	0	$\frac{1}{16}$	0	$\frac{5}{2}$	0	1			
6	0	0	1	0	5	0	1		
7	0	$\frac{1}{64}$	0	$\frac{91}{16}$	0	$\frac{35}{4}$	0	1	
8	0	0	1	0	21	0	14	0	1

Table 2. Central factorial numbers of the second kind $T(n,k)$. See also the sequences A395862, and A370703 in the OEIS [11].

5. Conclusions

In this manuscript, we derive closed formulas for multifold sums of powers of integers by combining the central Newton interpolation formula with hockey-stick identities for binomial coefficients. We further provide Wolfram Mathematica programs for the efficient verification of the derived identities.

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Mathematica Programs

The Mathematica programs in the project repository validate the results listed below.

Mathematica Program	**Result Validated**
`ValidateOrdinarySumsOfPowers.txt`	Proposition (3.2)
`ValidateDoubleSumsOfPowers.txt`	Proposition (3.4)
`ValidateMultifoldSumsOfPowers.txt`	Theorem (3.6)
`ValidateMultifoldBinomialSumsOfPowers.txt`	Proposition (3.7)
`ValidateOrdinarySumsOfPowersAtZero.txt`	Proposition (4.5)
`ValidateDoubleSumsOfPowersAtZero.txt`	Proposition (4.6)
`ValidateMultifoldSumsOfPowersAtZero.txt`	Theorem (4.7)
`ValidateMultifoldSumsOfPowersAtZeroBinomialForm.txt`	Proposition (4.8)

Metadata

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- **Web Version:** kolosovpetro.github.io/sums-of-powers-central-differences/
- **Sources:** github/kolosovpetro/SumsOfPowersCentralDifferencesNewtonFormula
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